This summer, I would like to work on robot learning in the real world through Prof. Andrea Bajcsy's Intent Lab. Before my interest in robotics, I applied to Penn wanting to improve the way people interacted with algorithms on the internet. My ambition was colored by my experience online during the pandemic. I spent significant time on platforms such as YouTube and Instagram, their recommendation systems created what I would now call a feedback control loop over much of my life. Thinking that a career in engineering or policy would help me make an impact on better "aligning" these systems, I thrust myself into a slew of coursework from machine learning to internet law. Through my experience, I realized that research would provide me the best opportunity to work on technical problems while building a platform for impact.

As industry labs push AI systems through the "scale is all you need" phase of deep learning, I have grown interested in two problems that might require a more principled approach: learning dynamics and safety in data-driven systems. While the problems are applicable generally to learning control systems, I currently find the domain of robotics most exciting to work on. Thinking about solutions, traditional controls have long motivated learning techniques, so I'm curious about what juice there is left to squeeze from the field's safety-critical tools. In particular, Prof. Andrea Bajcsy has been enthusiastic about using latent dynamics or "world models" as the canvas to translate these tools into the deep learning setting. Beyond its promise, I'm excited about this direction because it ties together my interests in representation learning, generative models, and controls. For these reasons, I believe a summer in Prof. Bajcsy's lab would be fruitful for my development as a researcher and roboticist.

My interest in representation learning grew out of a theoretical project in Prof. Nikolai Matni's group. Taking note of *The Bitter Lesson*, I wanted to learn about the current tools we have for search (optimization) and learning (optimization) over deep networks. Guided by grad student Thomas Zhang, I used my background from taking convex optimization to design and test second-order optimizers for representation learning. The project reminded me that minimizing test error is not the only proxy for learning. Studying linear representations, we used the subspace distances to bridge theory to numerics. I also designed experiments akin to what would be done when in practice: assessing the performance of our representations by training on new tasks in new distributions. Thinking about more empirical work, I have been excited by projects such as *AnyCar to Anywhere* from Prof. John Dolan's group which uses the same framework to learn a cross-embodied driving dynamics model. These techniques are still nascent, but will help bring robot learning out of the lab into real application. Some related directions I might explore at RISS are methods to learn representations conducive to safety constraints. Specifically I am curious about adapting tools from *Learning Control Barrier Functions from Expert Demonstrations* by Robey et al. into latent dynamics models. A target application might be reinforcement learning in the real world, where exploring outside of a safe set might lead to disastrous consequences.

My first experience thinking about dynamics in latent space though, began with my project on diffusion models in Prof. Eric Wong's lab during my sophomore summer. Reading *Diffusion Policy* and *AlphaFold 3*, I was curious about the success diffusion had in problem settings which were seemingly quite different from their origin in text-to-image generation. Listening to talks about these projects, the common thread seemed to be sequential decisions about local structure conditioned on global structure (e.g., planning trajectory curvature after choosing to go left or right around an obstacle). In line with the theme of the lab, my project approached this phenomenon from an explainability perspective. I wondered if an inference-time algorithm could uncover these sequential dependencies. To do so, I trained DDPMs on a couple

of datasets and analyzed their trajectories. Recalling that the final sample is just the end of a stochastic process, I computed its variance while conditioning on prefixes of the process. Setting aside technicalities, I noticed the variances for more "global" features would drop before others when conditioning on increasingly larger prefixes. This finding aligned with my hypothesis that DDPMs generate global structure earlier, but it left me with many more questions about the dynamics of the diffusion process. As the summer progressed, I attempted a few other explainability directions, yet I could not find a satisfying end to the story.

Still, generative models have since become a cornerstone in robotic manipulation, so I hope to leverage what I learned going forward. An idea that I have discussed with Prof. Matni is using the abstraction of flow-matching to generate physical motion plans from reduced-order dynamics models - allowing for planning directly in a world model's latent space. Flow-matching is a flexible framework, so there are many interesting questions regarding the best way to go about doing this. In preparing to apply, I spoke with Prof. Bajcsy, who mentioned that most of the robots in her lab already had good baselines for further work on diffusion-style policies. In particular, Abigail Defranco from the RISS 2024 cohort worked on classifier guidance to align diffusion policies during her time in the Intent Lab. In her project, the classifier's gradient was used as a control input over the diffusion policy's dynamics and aligned the final sample with a user's preferences. The way Abigail and Prof. Bajcsy reasoned about how the diffusion model's dynamics resonated with how I thought about them. During the conversation, Prof. Bajcsy also encouraged me to reach out to Prof. Simchowitz in the MLD, whose theoretical work on generative models might also serve as inspiration for a project. These experiences assured me that the Intent Lab would have the resources to support me this summer.

Beyond robotics, generative models, and machine learning, my primary intention with RISS is to become a better researcher. I have had the luxury of learning advanced theory in class, and programming complicated systems at work, but what really distinguishes a researcher from a student or engineer is the ability to find, motivate, and communicate important problems. Beyond the working paper, Prof. Bajcsy has helped students such as Hyun Joe Jeong prepare work for major conferences, and older RISS alumni have become frequent contributors to the field. I was pleasantly surprised to see that Wenli Xiao, the first author of *AnyCar to Anywhere* (mentioned above), actually began his work with Prof. John Dolan in the RISS 2022 cohort. During RISS he worked on safety in multi-agent reinforcement learning, leveraging techniques from game theory and control. Since then he has led and contributed to many projects in the RI. My understanding from these student's stories is that RISS is an incredible opportunity for me to leverage my technical skills and become an impactful member in the robotics community.

On this note, finding communities worth contributing to has always been a North Star of mine. School has given me much joy throughout the years, and I have found ways to give back through student government, peer advising and especially teaching. During my Freshman summer, I assisted one of Penn Engineering's high school programs. I was surrounded by students from 5 continents with a common interest in applied math and computer science. In less than four weeks, they bonded over stories, food, and *of course* my problem sets to become friends who are still in touch today. I feel that RISS may provide a similar environment for building relationships extending far beyond the program, so I will eagerly wait to find out if I may find a new home on CMU's campus this summer.